

Slide 1 – Title

AI-Based Smart Irrigation System Using IoT

Presented by: Your Name

Slide 2 – Introduction

Content

Agriculture is one of the most important sectors in the world. Water management is a major challenge due to climate change and increasing water scarcity. Traditional irrigation methods often waste water and require continuous manual monitoring. A Smart Irrigation System uses IoT sensors and automation to provide water only when crops need it, improving water efficiency and crop productivity.

Slide 3 – Abstract 1

Summary

This paper reviews IoT-based smart irrigation systems, including sensors, communication technologies, and monitoring techniques used for efficient water management.

Drawbacks

- No practical implementation.
 - No real-time field testing.
 - Literature review only.
-

Slide 4 – Abstract 2

Summary

This paper uses fuzzy logic with soil moisture, temperature, and humidity sensors to automate irrigation using GSM communication.

Drawbacks

- Depends on GSM network.
 - Manual calibration required.
 - Higher communication cost.
-

Slide 5 – Abstract 3

Summary

This system uses Arduino and the ThingSpeak cloud platform for remote irrigation monitoring and control through smartphones.

Drawbacks

- Requires stable internet.
 - Cloud service limitations.
 - Limited performance validation.
-

Slide 6 – Abstract 4

Summary

This paper explains how smart irrigation improves water-use efficiency and supports Sustainable Development Goal (SDG 6).

Drawbacks

- Review paper only.
 - No practical implementation.
 - No experimental validation.
-

Slide 7 – Abstract 5

Summary

This paper discusses the role of IoT-based smart irrigation in sustainable agriculture and efficient water management.

Drawbacks

- No real-time testing.
 - Depends on previous studies.
 - No quantitative results.
-

Slide 8 – Existing System Drawbacks

Common Drawbacks

- Internet/GSM dependency.
 - Lack of real-time field validation.
 - No AI-based decision making.
 - No crop disease detection.
 - No weather prediction.
 - High maintenance and monitoring effort.
-

Slide 9 – Proposed New Project

AI-Based Smart Irrigation System Using IoT

Content

To overcome the limitations of previous systems, we propose an **AI-Based Smart Irrigation System**. The system uses soil moisture, temperature, humidity, and rainfall sensors to monitor field conditions. AI analyses the collected data and automatically controls irrigation. Farmers can monitor the field through a mobile application and receive real-time alerts.

New Features

- AI-based irrigation decision making.
- Automatic water pump control.
- Real-time IoT monitoring.
- Weather prediction integration.
- Water-saving algorithm.

- Mobile notifications.
 - Low water consumption.
 - Improved crop yield.
-

Slide 10 – Advantages

- Saves water.
 - Reduces labour.
 - Increases crop productivity.
 - Automatic irrigation.
 - Remote monitoring.
 - Cost-effective farming.
 - Environment friendly.
-

Slide 11 – Conclusion

The proposed **AI-Based Smart Irrigation System** overcomes the limitations of existing irrigation systems by integrating AI, IoT, and automation. It improves water management, reduces manual effort, and supports sustainable agriculture with better crop productivity.